

# Smart Energy Meter with IoT Billing

Team NEXUS 4

Ishan Jiji George

Manuel James

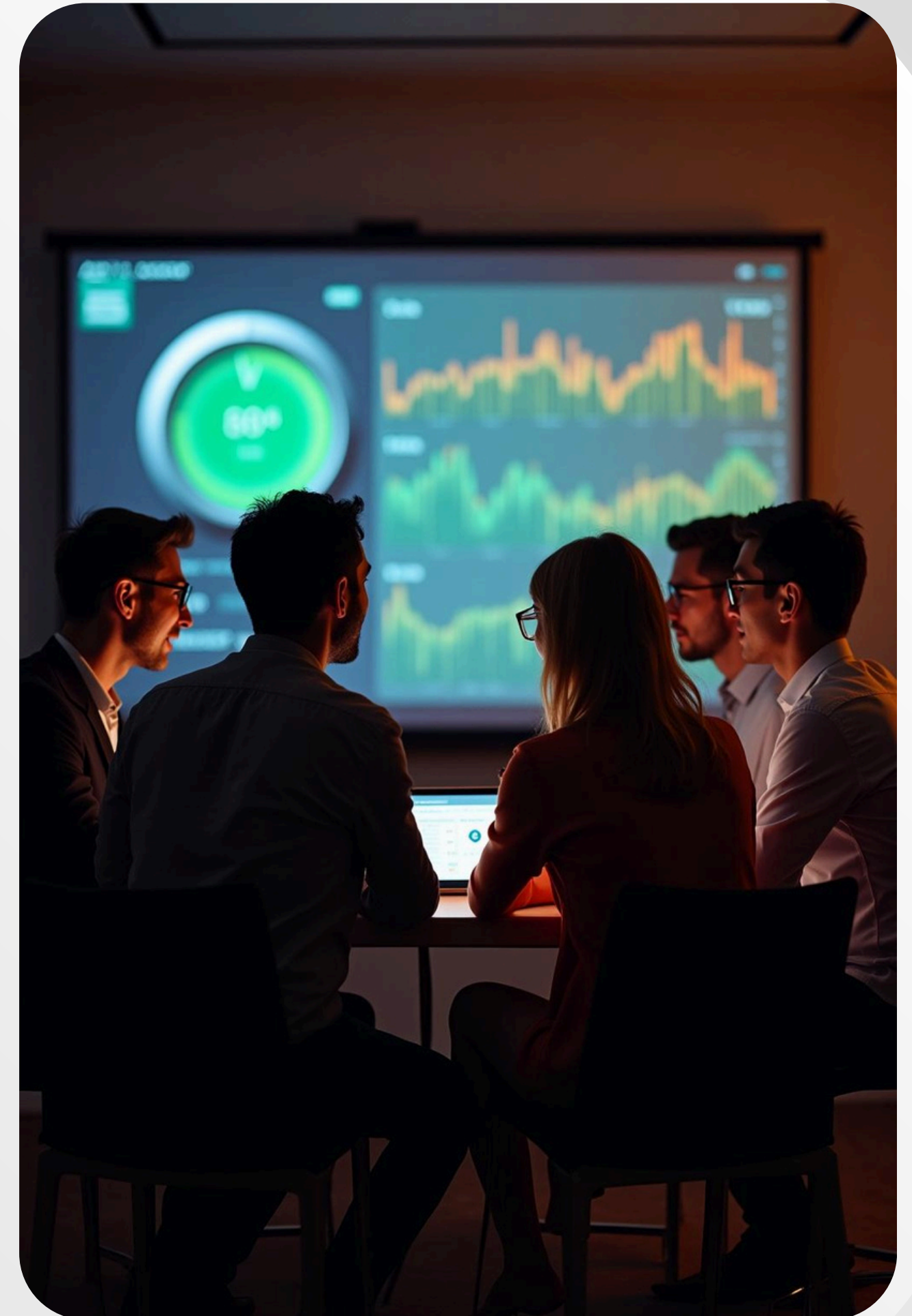
Sherry John

Sona Salu



# Introduction

- Combines smart meters with IoT for efficient energy management.
- Uses ESP8266 NodeMCU and sensors for real-time monitoring.
- Enables remote control and live tracking of power usage.



# Overview of Smart Energy Meters

- Tracks real-time electricity usage for accurate energy management.
- Replaces traditional meters that require manual reading and physical billing.
- Calculates power and energy consumption using current sensors and microcontrollers.
- Supports residential, commercial, and industrial applications with scalability and ease of integration.

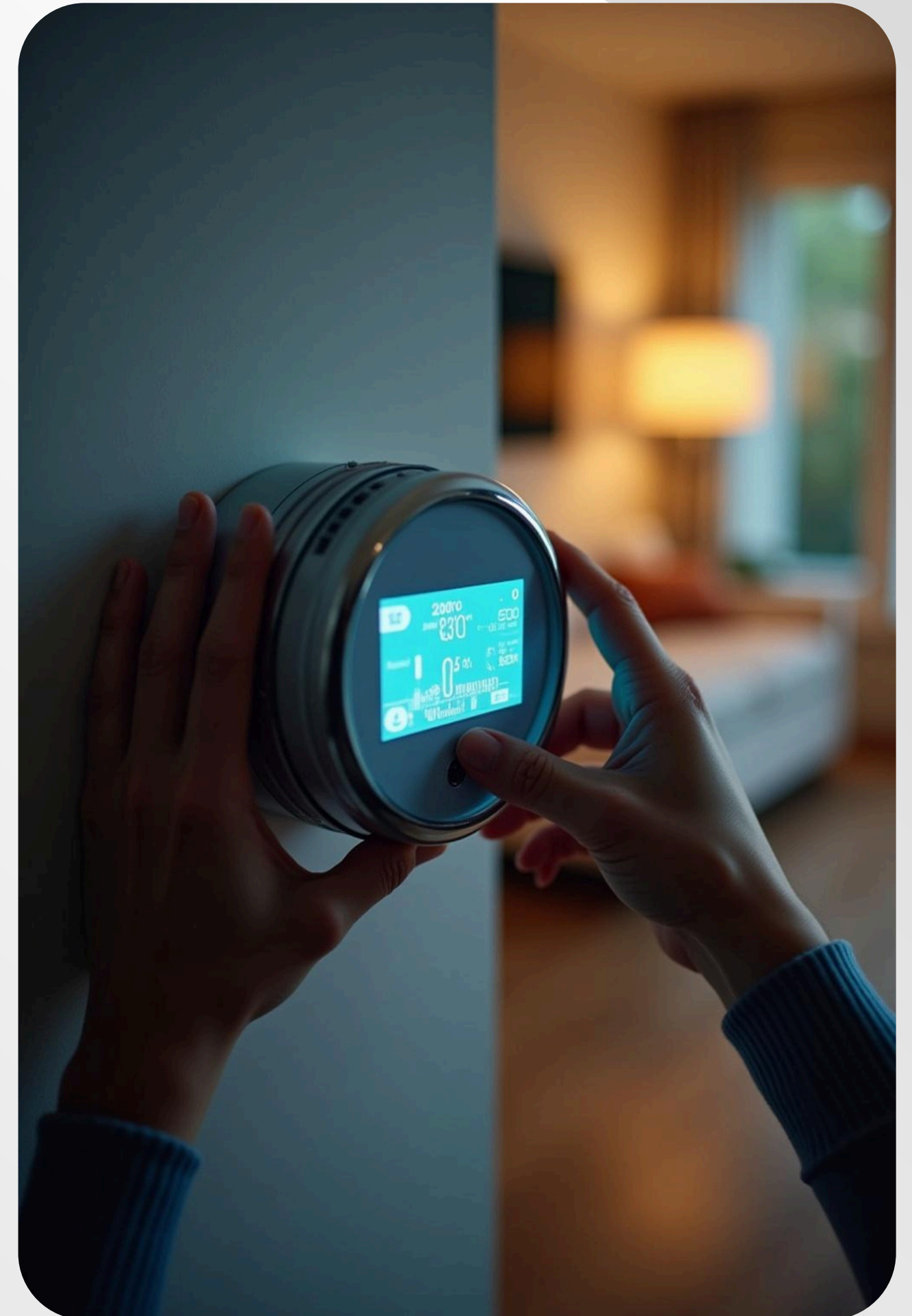
# Importance of IoT in Energy Management

- Enables real-time data transmission to the cloud via Wi-Fi (ESP8266).
- Provides remote access to energy usage through mobile apps or web dashboards.
- Automates billing and notifications using cloud platforms and email APIs.
- Supports integration with smart grid infrastructure for better load management and forecasting



# Key Benefits of Implementing Smart Energy Meters

- Eliminates manual billing errors through automated and precise calculations.
- Improves user awareness of consumption, encouraging energy-saving behavior.
- Reduces operational costs for utility providers and consumers alike.
- Promotes digital transformation and paperless billing with real-time alerts and logs.



# Key Components

- ESP8266 NodeMCU
- AD1115 Module
- SCT-013 100A/1V Current Sensor
- ZMPT101B Single Phase Voltage Sensor
- 5V 1 channel relay
- BC547, 1K resistor, 10K resistor and 0.1uf capacitor
- Breadboard and jumperwires
- Firebase
- Google Script

# Implementation Plan

## HARDWARE SETUP

- Connect the current sensor (CT/Shunt) to the ADS1115 ADC module.
- Interface ADS1115 with ESP8266 NodeMCU using I2C communication.
- Ensure proper power supply (5V) to all components.

## FIRMWARE DEVELOPMENT

- Write code in Arduino IDE to:
- Read current values from ADS1115.
- Calculate power and energy consumption.
- Format data for cloud upload and billing.

## CLOUD INTEGRATION

- Set up a Firebase Real-time Database to store energy usage data.
- Integrate Firebase with the ESP8266 using appropriate libraries.
- Enable real-time data sync between the device and Firebase

## BILLING & EMAIL AUTOMATION

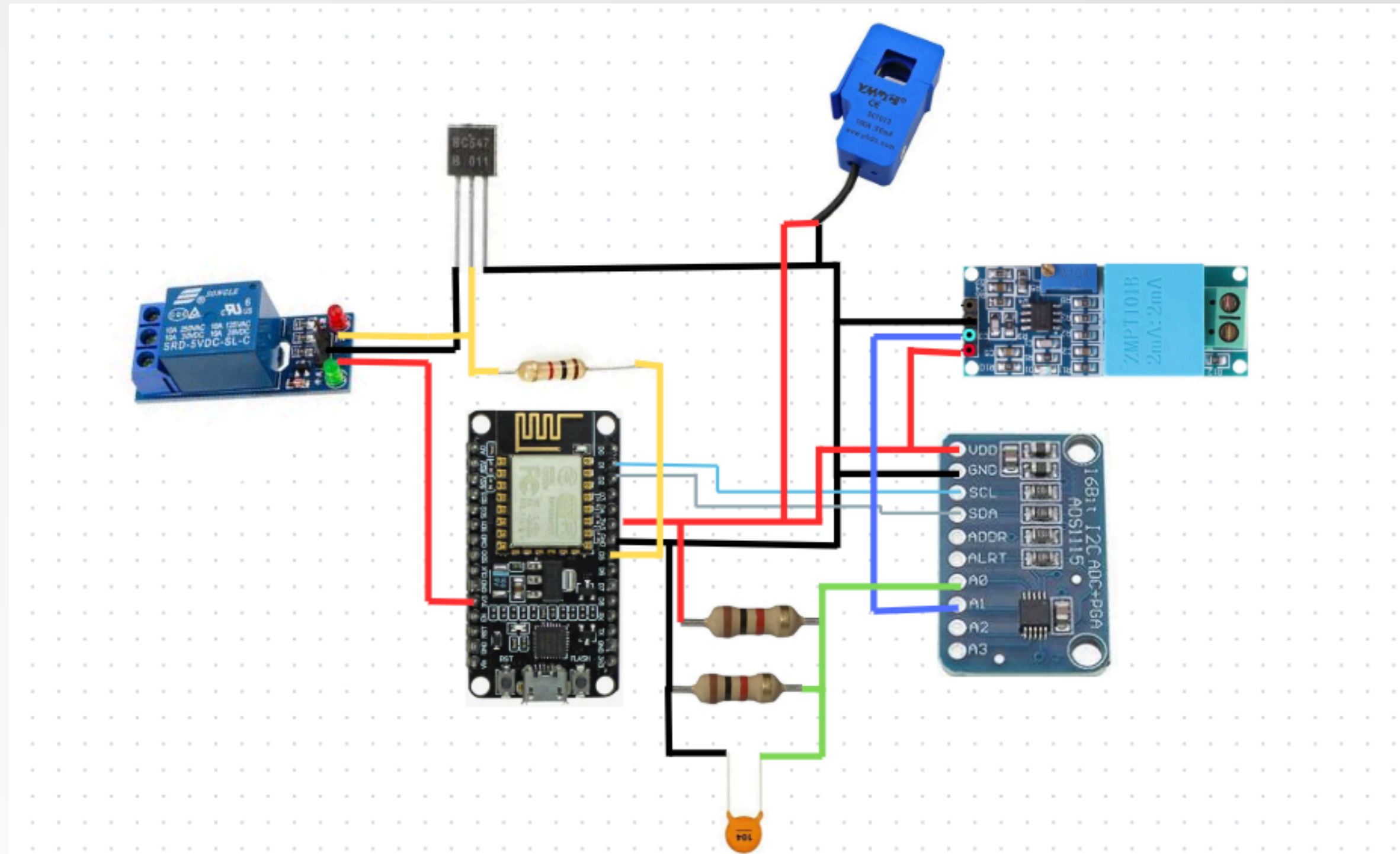
- Implement monthly billing logic based on total energy consumed.
- Use SMTP2GO, SendGrid, or IFTTT to configure email API services.
- Automatically send bill summaries to user emails at the end of each billing cycle

## TESTING AND VALIDATION

- Simulate different load conditions to test accuracy.
- Verify data upload, bill generation, and email delivery.



# Circuit Diagram



# Working Principle

## Current Measurement

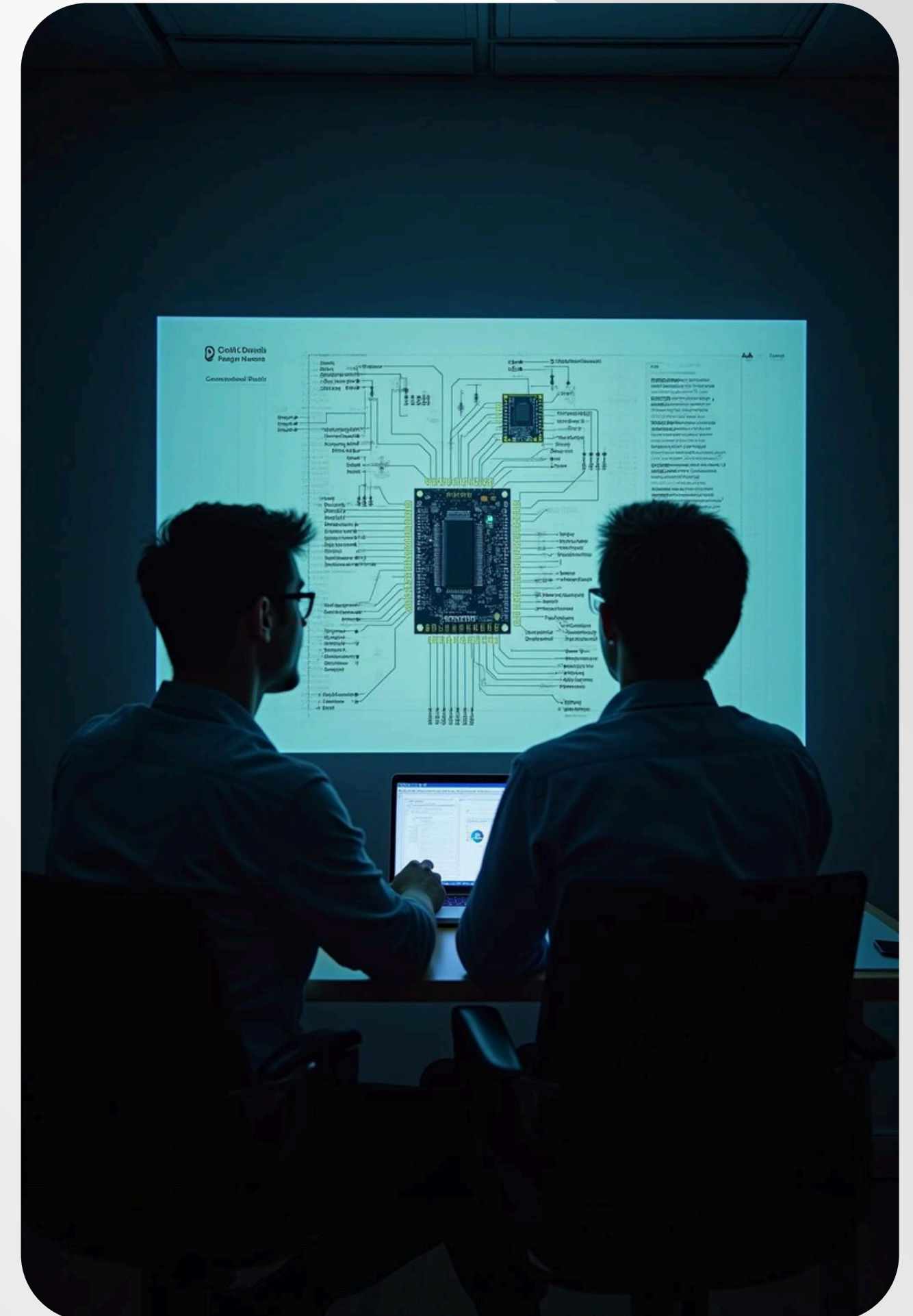
A CT sensor or shunt resistor detects the current flow in the electrical circuit.

## Analog-to-Digital Conversion

The ADS1115 ADC module converts the small analog voltage (from the sensor) into a precise digital value.

## Data Processing

The ESP8266 NodeMCU calculates power and energy consumption using the digital current data and a fixed voltage reference.





## Cloud Upload

The ESP8266 uploads real-time energy data to Firebase using Wi-Fi.

## Billing Logic

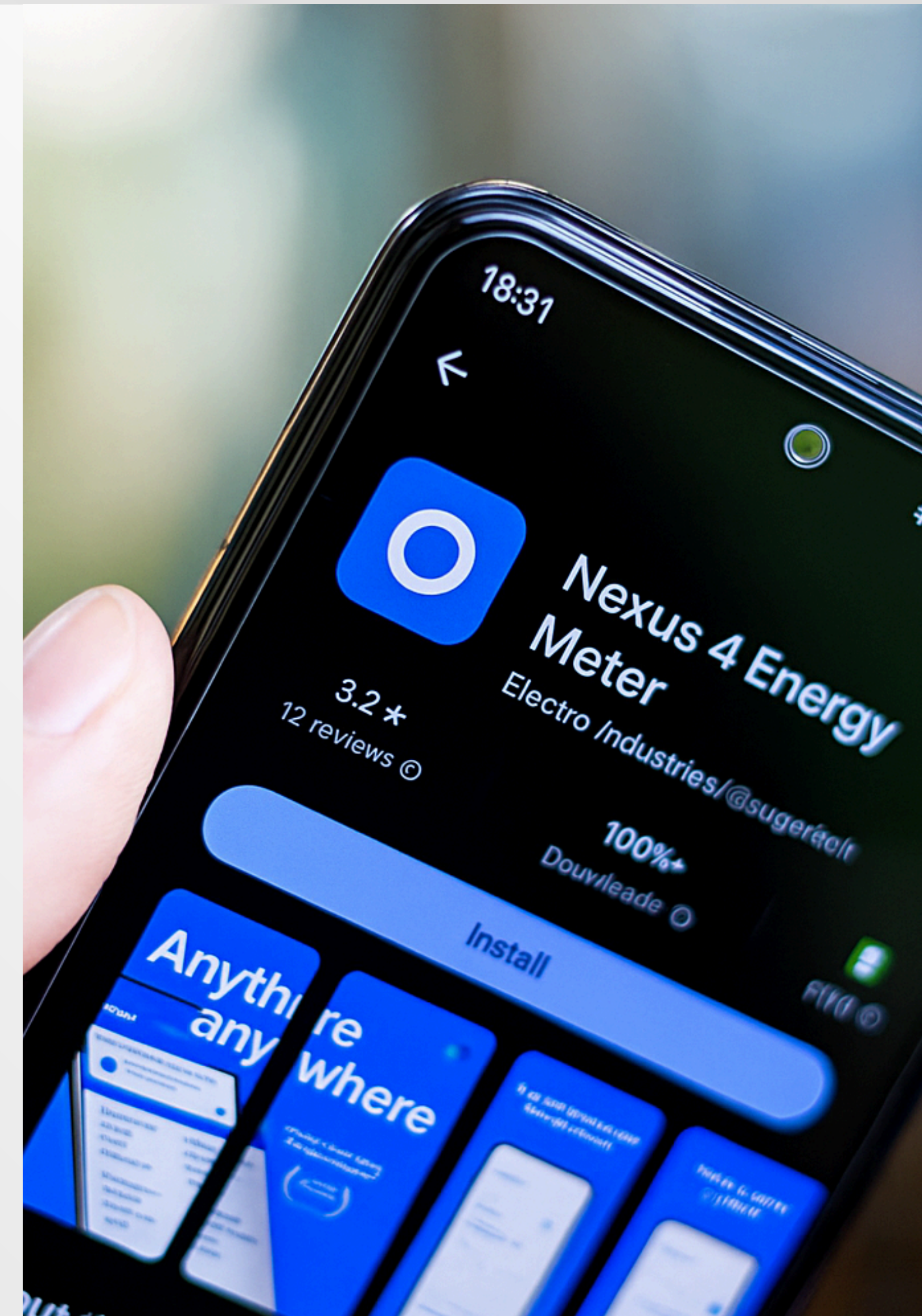
The system calculates the total energy consumed monthly and applies a unit rate to compute the bill amount.

## Email Automation

The final bill is sent to the user's email via SMTP2GO, SendGrid, or IFTTT automatically at the end of each billing cycle.

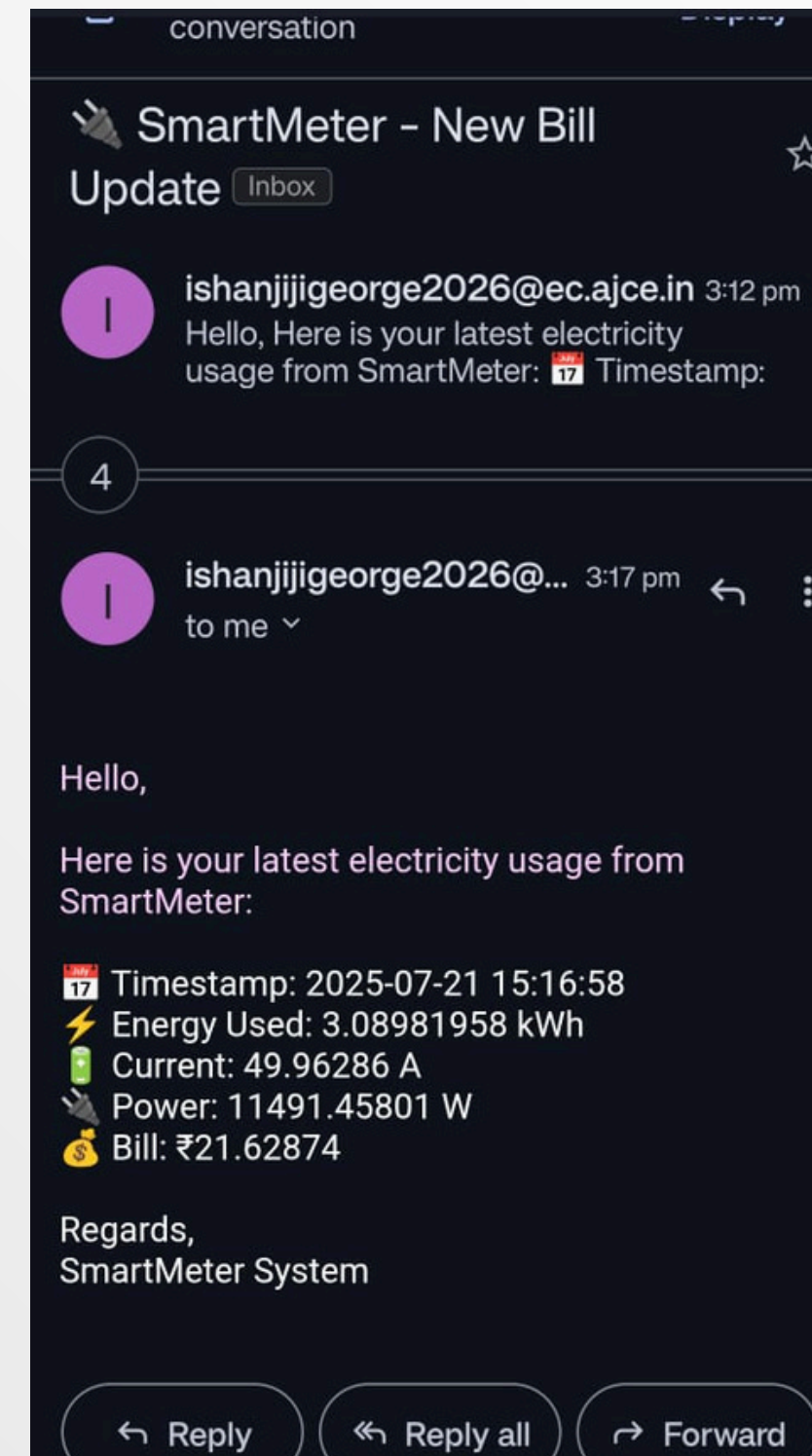
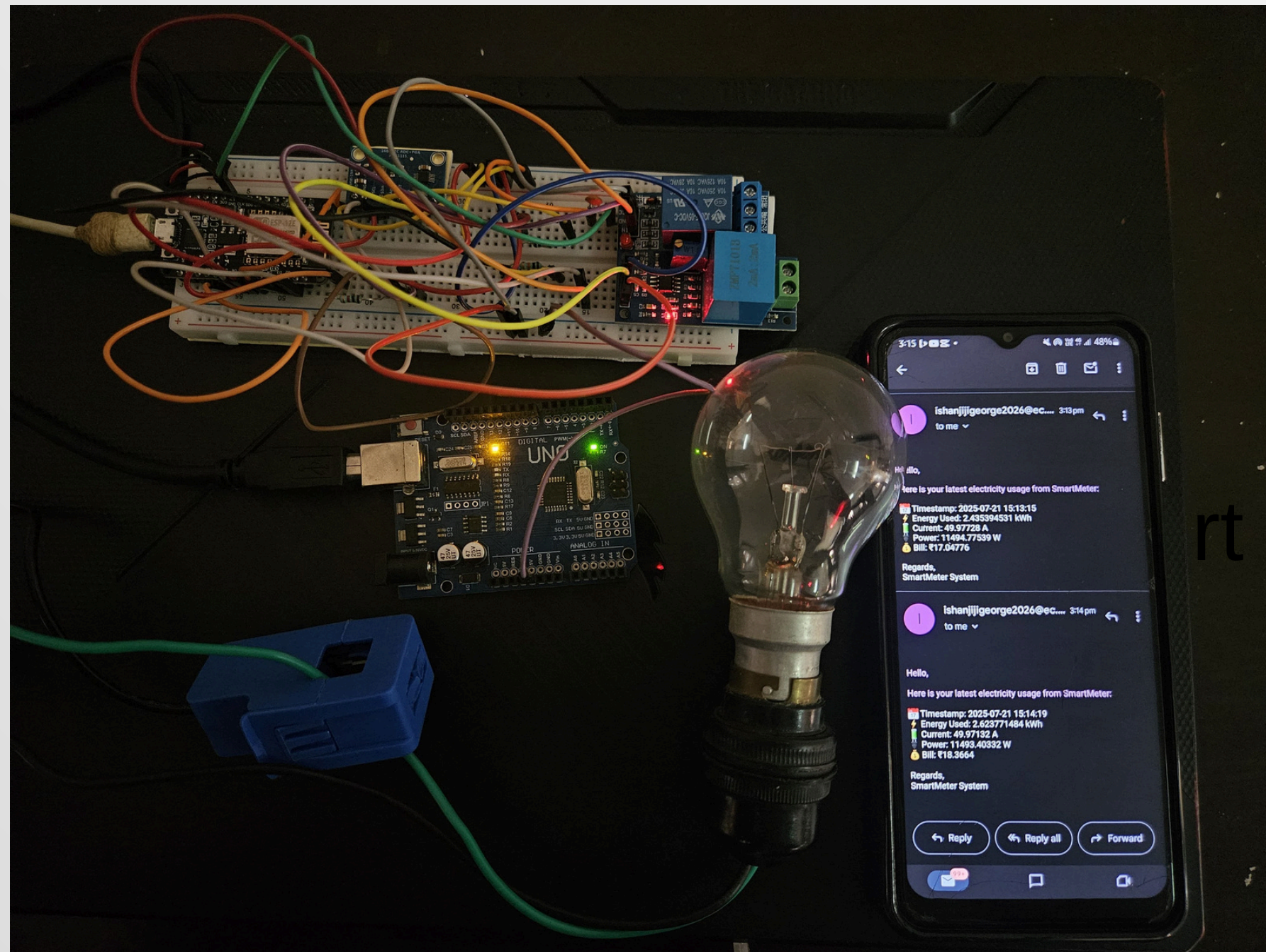
## User Monitoring

Users can monitor their live and historical energy usage from any device via the cloud.





# Result





# Advantages

- Real-Time Monitoring
- Automated Billing
- Remote Access
- High Accuracy
- Cost and Time Efficient
- Eco-Friendly
- Scalable Design
- User Awareness

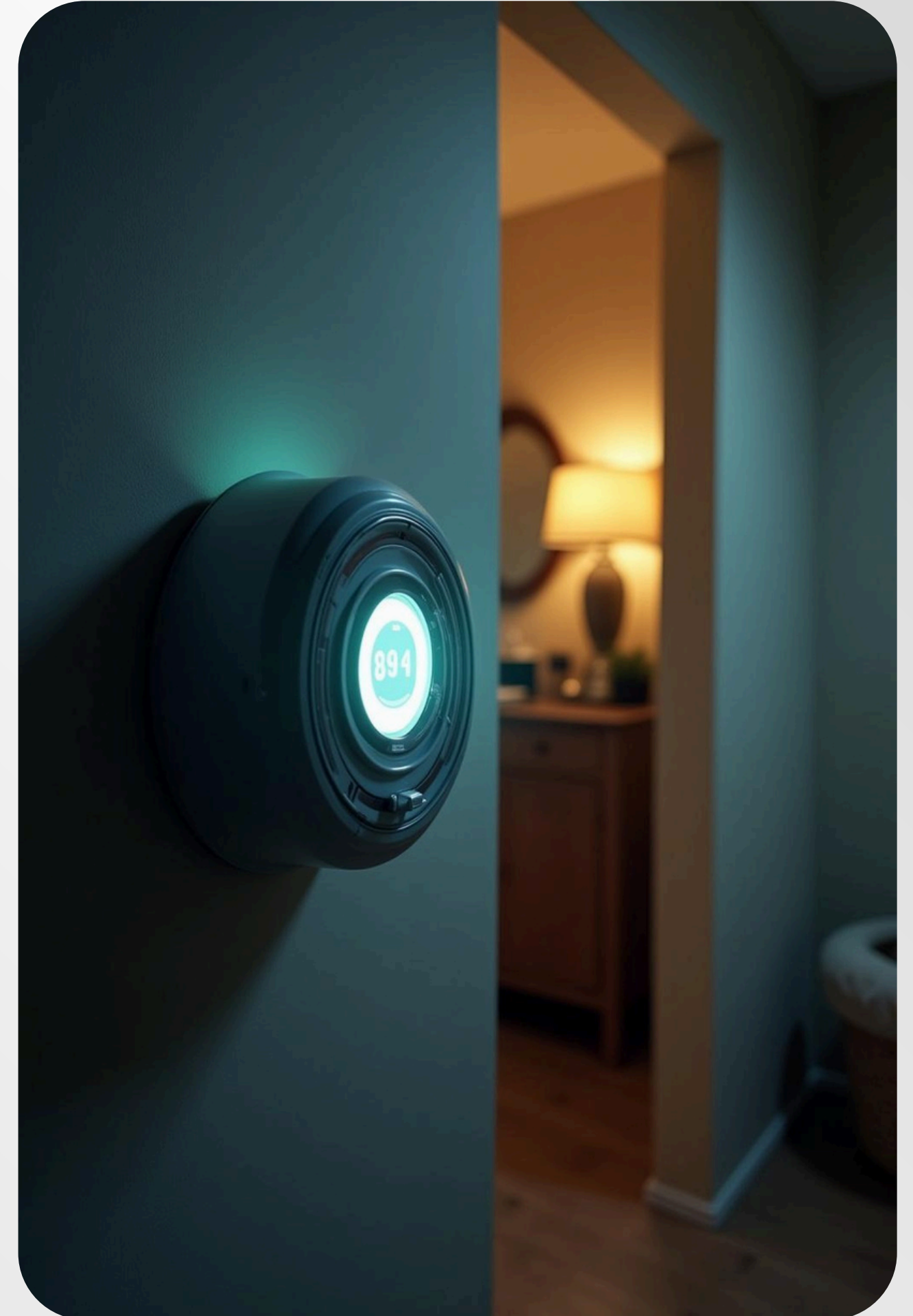
# Future Scope

- Dual Monitoring: Voltage + Current
- Mobile App Integration.
- Load Control Automation.
- Dynamic Tariff Billing
- Solar Integration



# Conclusions

- Revolutionizes Energy Management
- Efficient & User-Friendly System
- Promotes Energy Conservation
- Future-Ready and Scalable





# Thank you!

Do you have any questions?

TEAM NEXUS 4